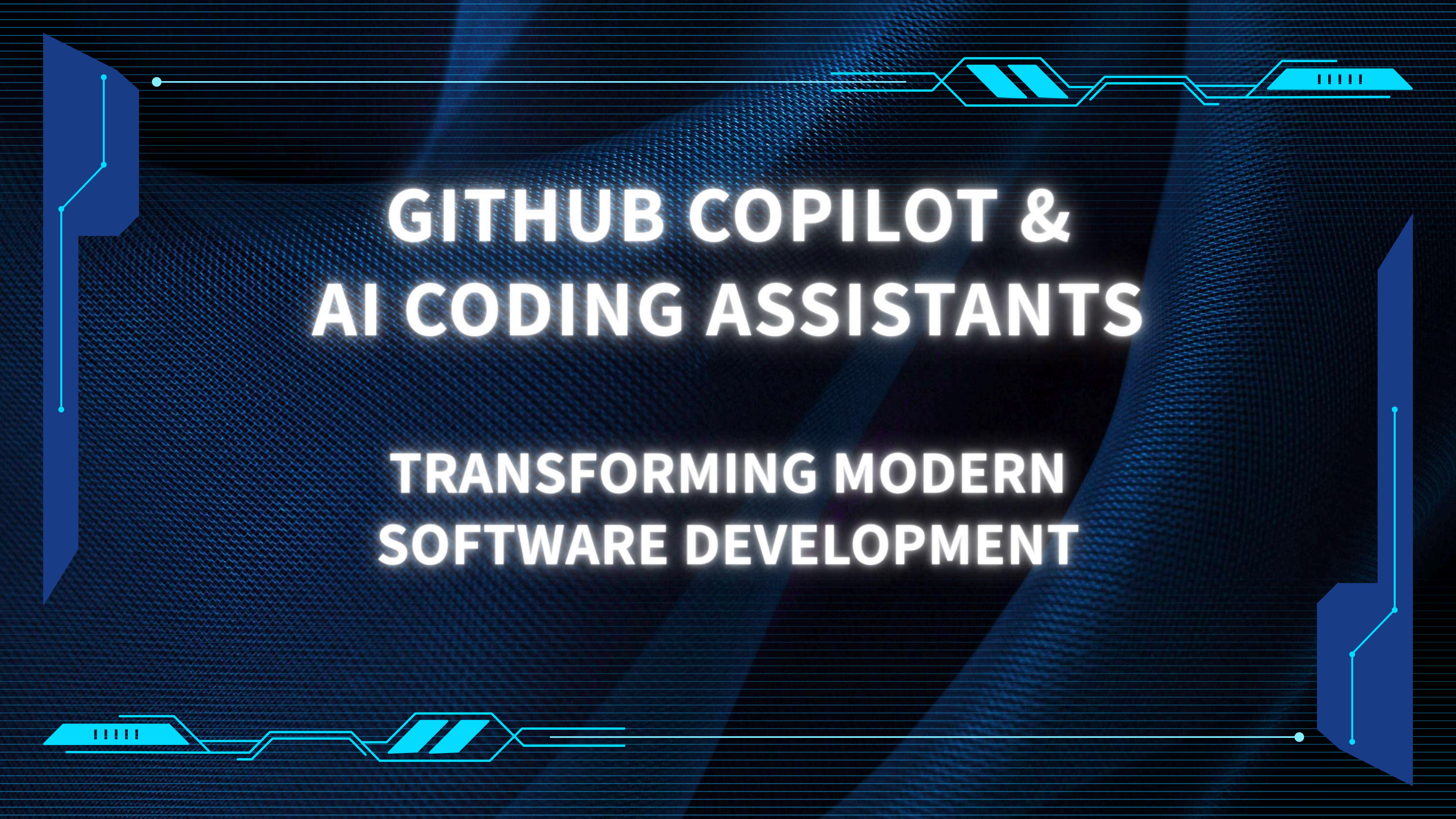
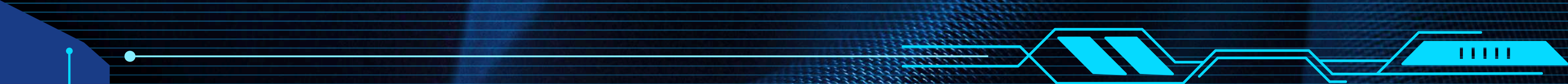




GITHUB COPILOT & AI CODING ASSISTANTS

TRANSFORMING MODERN SOFTWARE DEVELOPMENT



OUR TEAM

Seerat Kaur
Japneet Kaur
Idowu Adesoye
Kyaw Khine Khant

WHAT WE'LL COVER

- 1. What is GitHub Copilot?
- 2. How AI Coding Assistants Work
- 3. Key Features
- 4. Benefits and Limitations
- 5. Real-World Use
- 6. Future Trends and Conclusion

INTRODUCING GITHUB COPILOT

2021 — Released by GitHub and OpenAI
AI-powered assistant inside VS Code
Suggests and completes code in real time
Supports multiple programming languages

CORE FEATURES OF GITHUB COPILOT

- **Code Completion**
Predicts and completes code as you type
- **Chat Interface**
Ask questions in plain language.
- **Code Explanation**
Breaks down complex logic clearly
- **Debugging Assistance**
Finds and fixes errors automatically.
- **Test Generation**
Creates unit tests instantly
- **Code**

UNDER THE HOOD: HOW IT WORKS

- Step 1: Reads your code and comments
- Step 2: Analyzes patterns and structure
- Step 3: Suggests next lines or full functions
- Step 4: Developer accepts, edits, or ignores
- Step 5: Model improves via retraining
- "Does not learn from your code in real time."

WHY DEVELOPERS LOVE AI ASSISTANTS

- Faster coding with fewer errors
 - Beginner-friendly and accessible
 - Anticipates your next line of code
 - Catches bugs and suggests fixes
 - Auto-generates comments and documentation
- "AI assistants help developers focus on logic, not syntax."

LIMITATIONS TO BE AWARE OF

- ⚠ May Generate Buggy Code
Always review AI output before using in production.
- ⚠ Depends on Training Data
Weaker performance for niche or uncommon languages.

RISKS AND ETHICAL CONCERNS

- ⚠ Security Risks
May generate vulnerable code.
- ⚠ Privacy Concerns
Data could be exposed.
- ⚠ Copyright Issues
May reproduce licensed code.
- ⚠ Incorrect Code
Bugs that look convincing.
- ⚠ Overreliance
Weakens critical thinking.
- ⚠ Skill Stagnation
Slows developer growth.

GITHUB COPILOT IN PRACTICE

- Write code faster.
- Solve programming problems.
- Generate automated tests.
- Write documentation.
- Learn new languages on the fly.

WHAT'S NEXT FOR AI CODING TOOLS

- Today
 - More accurate suggestions
 - Bug detection
 - Code explanation
- Future
 - Industry-wide adoption
 - Full codebase understanding
 - Automated testing & docs
 - AI as standard dev tool

KEY TAKEAWAYS

- 1. AI assistants are reshaping software development.
 2. GitHub Copilot boosts productivity.
 3. Supports beginners and experts alike.
- 1. Human review remains essential.
 2. AI will become a standard developer tool.

"AI assists. Humans decide."

REFERENCES

GitHub, Inc. (2023). GitHub Copilot documentation. GitHub Docs.
<https://docs.github.com/en/copilot>

Chen, M., et al. (2021). Evaluating large language models trained on code. OpenAI Research.
<https://arxiv.org/abs/2107.03374>

OpenAI. (2023). OpenAI Codex: Technical overview and capabilities. OpenAI Research Papers.
<https://openai.com/research>

Course Materials. (2024). Introduction to AI-assisted software development. Course Repository /
Institution LMS.

Tabassian, M., & Zagalsky, A. (2022). AI coding assistants in practice: Benefits and risks. IEEE
Software, 39(4), 12–18.



THANK
YOU